

The Regime Integration Gradient (v_{RIG}): A Fundamental Velocity Scale for Consciousness as 2D→3D Information Transfer

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[Johann]: Kann ich das so hochladen?

Abstract

We derive a fundamental velocity scale $v_{\text{RIG}} \approx 1,352 \text{ km/s}$ from first principles, representing the characteristic rate at which consciousness integrates two-dimensional holographic information into three-dimensional volumetric perception. This velocity emerges geometrically from the product of two universal constants: the inverse fine-structure constant $\alpha^{-1} \approx 137.036$ (governing electromagnetic transparency) and the golden ratio $\Phi \approx 1.618$ (governing optimal spatial scaling). The resulting formula $v_{\text{RIG}} = c/(\alpha^{-1} \cdot \Phi)$ links quantum electrodynamics, fractal geometry, and consciousness studies within a unified framework. We propose testable predictions relating metabolic scaling (Kleiber's Law), critical flicker frequency, and neuronal integration timescales to this fundamental constant. Our framework suggests consciousness operates as a **regime integration process**, converting surface-entropy-governed photon propagation (2D, $S\emptyset A$) into volume-entropy-governed experiential moments (3D, $S\emptyset V$) at a rate fundamentally limited by v_{RIG} .

Keywords: consciousness, holographic principle, fine-structure constant, golden ratio, metabolic scaling, information integration, entropy governance, OIPK framework

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